

Practice Problems

Question 1

Your friend is deciding between continuing to use a gasoline scooter or switching to an electric bike for their daily commute. The scooter consumes fuel at a rate of 35 km/l, which runs on gasoline, and the energy content of gasoline remains 10 kWh/l. The electric bike uses 0.05 kW h/km. Which mode of transport uses less energy daily and by how much, considering that the daily commute is 25 km %?

Question 2

Two friends are comparing the energy efficiency of their vehicles for a 100 km weekend trip:

- Sam drives an electric vehicle (EV) that consumes 0.18 kWh/km.
- Alexa drives a hydrogen car that consumes 1.1 kg of hydrogen per 100 km.
- The energy content of hydrogen is 33.3 kWh/kg.

- (a) Calculate the total energy used by each vehicle for the 100 km trip.
- (b) Which vehicle is more energy-efficient, and by how many percent?

Question 3

A regional transport agency is evaluating whether to replace a short-haul domestic flight route (300 km) between two coastal cities in New Zealand with a high-speed electric ferry. Each mode carries 200 passengers per trip.

You are given the following information:

Parameter	Domestic Aircraft	High-speed Ferry
Energy consumption (kWh/km)	20	15
Passenger capacity	200	200
Distance	300 km	300 km

- Distance per trip: $d = 300$ km
- Passenger count: $N = 200$

- Aircraft energy consumption: $e_{\text{air}} = 20 \text{ kWh/km}$
- Ferry energy consumption: $e_{\text{ferry}} = 15 \text{ kWh/km}$

- Calculate the energy consumption per passenger-km for both the aircraft and the ferry.
- Determine the total energy required per trip for each mode.
- How much energy is saved per trip by switching to the ferry?
- If this trip occurs twice daily for a full year, how much energy is saved annually?
- If the fuel costs 100 NZD/MWh, how much is the society saving on fuel annually? If the ferry takes 15 min longer than the plane, and if the average person's time is worth 40 NZD/h, how much person-hours are lost in a year, and what is the cost of that time?

Question 4

A regional train in the South Island transports on average 180 passengers per trip over a distance of 100 km. The electric train consumes 12 kWh/km. What is the energy consumption in kWh/person-km?

Question 5

A small commuter van (diesel) is used for airport shuttle services. It has a fuel efficiency of 10 km/l and a seating capacity of 10 passengers. The diesel energy content is 10 kWh/l. If the van operates at full capacity over a 200 km trip, calculate the energy consumption per person-km.

Question 6

Christchurch's electric bus fleet carries approximately 35 passengers per trip on average. Assume an electric bus travels 300 km per day and consumes 0.5 kWh/km.

- Calculate the daily person-km transported by one electric bus.
- Determine the energy consumption per person-km in kWh.
- Compare this value to a single-occupancy gasoline car that travels 14,000 km/year, with a fuel efficiency of 12 km/l and an energy content of 10 kWh/l. What is the energy consumption per person-km for the car?
- Which mode is more energy-efficient per person-km, and by what factor?

Question 7

A fully loaded electric delivery van accelerates from 30 km/h to 90 km/h in 12 s while merging onto a motorway. The total energy required for this acceleration, including kinetic energy, rolling resistance, and aerodynamic drag, is:

$$E_{\text{total}} = 620 \text{ kJ}$$

Assume the vehicle travels on a flat road with constant acceleration.

The van has:

$$C_{\text{roll}} = 0.012$$

$$C_d = 0.28$$

$$A = 2.8 \text{ m}^2$$

- (a) Calculate the mass of the vehicle.
- (b) Determine the percentage contribution of:
 - kinetic energy,
 - rolling resistance,
 - aerodynamic dragto the total energy consumption.
- (c) Briefly explain why aerodynamic drag becomes increasingly important at higher vehicle speeds.